

**SIMATS ENGINEERING  
ASSESSMENT TOOL 4**

**COURSE NAME: Artificial Intelligence**

**COURSE CODE: CSA17**

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**COURSE OUTCOME COVERED**

***CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

*Assessment Tool Weightage: 20%*

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**QUESTIONS: 5**

**Time Duration: 1 Hour  
Total Marks:30**

Annexure A

**Resolution Algorithm Implementation**

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***Code of Conduct:***

*I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

## Annexure A

### Resolution Algorithm Implementation

#### Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

#### Goal

Prove that **the ground is wet** using the Resolution Algorithm.

#### Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

**Negate the goal.**

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ( $\square$ ) if the goal is proved.

#### Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

#### Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

**Negate the goal.**

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

#### Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

#### Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

#### Question 4 – Placement Eligibility

##### Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

##### Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

##### Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

#### Question 5 – Access Control System

##### Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

##### Goal

Prove that **the user is granted access** using the Resolution Algorithm.

##### Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ( $\square$ ).
5. Explain each resolution step and write the final conclusion.

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#### Question 1 – Rain and Wet Ground

##### 1. Propositional Logic

Let:

- **R** = It is raining.
- **W** = The ground is wet.

Knowledge Base:

1. If it rains, then the ground becomes wet:  
 $R \rightarrow W$
2. It is raining:  
**R**

Goal:

**W**

##### 2. CNF Conversion

Implication:

$$R \rightarrow W \equiv \neg R \vee W$$

So the clauses are:

- **C1:**  $\neg R \vee W$
- **C2:** **R**

##### 3. Negate the Goal

Goal = **W**

Negated goal:

**$\neg W$**

Add it to the clause set:

- $C1 = \neg R \vee W$
- $C2 = R$
- $C3 = \neg W$

#### 4. Resolution Steps

Resolve  $C1$  and  $C2$ :

**$(\neg R \vee W), (R)$**

Resolving **R** and  **$\neg R$** :

**W**

New clause:

**C4: W**

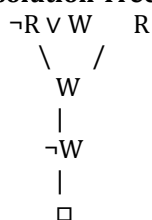
Now resolve  $C4$  with  $C3$ :

**$(W), (\neg W)$**

Therefore:

□ **(Empty Clause)**

#### Resolution Tree



#### Conclusion

Since the **Empty Clause** (□) is derived, the goal is proved.

**Therefore, the ground is wet.**

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### Question 2 – Student Assignment Submission

#### 1. Propositional Logic

Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

Knowledge Base:

1. If Rahul submits the assignment, then Rahul receives internal marks:

**$S \rightarrow M$**

2. Rahul submitted the assignment:

**S**

Goal:

**M**

#### 2. CNF Conversion

**$S \rightarrow M \equiv \neg S \vee M$**

Clauses:

- **C1:  $\neg S \vee M$**
- **C2: S**

#### 3. Negate the Goal

Goal = **M**

Negated goal:

**$\neg M$**

Add:

- **C3 =  $\neg M$**

#### 4. Resolution

Resolve  $C1$  and  $C2$ :

**$(\neg S \vee M), (S)$**

Resolving **S** and  **$\neg S$** :

**M**

New clause:

**C4: M**

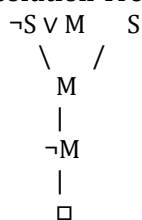
Now resolve  $C4$  and  $C3$ :

**$(M), (\neg M)$**

Therefore:

□

### Resolution Tree



### Conclusion

The Empty Clause □ is derived.

**Therefore, Rahul receives internal marks.**

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## Question 3 – Library Membership

### 1. Propositional Logic

Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

Knowledge Base:

1. If a person is a library member, then the person can borrow books:  
 **$L \rightarrow B$**
2. Priya is a library member:  
 **$L$**

Goal:

**B**

### 2. CNF Conversion

**$L \rightarrow B \equiv \neg L \vee B$**

Clauses:

- **C1:  $\neg L \vee B$**
- **C2:  $L$**

### 3. Negate the Goal

Goal = **B**

Negated goal:

**$\neg B$**

Therefore:

- **$C3 = \neg B$**

### 4. Resolution

Resolve:

**$C1 = \neg L \vee B$**

with

**$C2 = L$**

Resolving  $L$  and  $\neg L$  gives:

**$C4 = B$**

Now resolve:

**$C4 = B$**

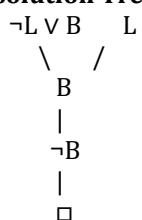
with

**$C3 = \neg B$**

Therefore:

□

### Resolution Tree



### 5. Final Conclusion

The Empty Clause □ is obtained.

**Therefore, Priya can borrow books.**

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## Question 4 – Placement Eligibility

### 1. Propositional Logic

Let:

- **A** = Arun cleared the aptitude test.
- **E** = Arun is eligible for placement.

Knowledge Base:

1. If Arun clears the aptitude test, then Arun is eligible for placement:

$$A \rightarrow E$$

2. Arun cleared the aptitude test:

$$A$$

Goal:

$$E$$

### 2. CNF Clauses

Convert the implication:

$$A \rightarrow E$$

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Therefore:

$$\neg A \vee E$$

Clauses:

- **C1:**  $\neg A \vee E$
- **C2:**  $A$

### 3. Negate the Goal

Goal:

$$E$$

Negated goal:

$$\neg E$$

Add:

- **C3:**  $\neg E$

### 4. Resolution Steps

Resolve C1 and C2:

$$(\neg A \vee E), (A)$$

Resolving A and  $\neg A$ :

$$E$$

Therefore:

$$\mathbf{C4: E}$$

Now resolve C4 and C3:

$$(E), (\neg E)$$

Therefore:

□

### 5. Resolution Tree

$$\begin{array}{c} \neg A \vee E \quad A \\ \backslash \quad / \\ E \\ | \\ \neg E \\ | \\ \square \end{array}$$

### Conclusion

Since the **Empty Clause** (□) is derived, the goal is proved.

**Therefore, Arun is eligible for placement.**

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## Question 5 – Access Control System

### 1. Propositional Logic

Let:

- **P** = User enters the correct password.
- **A** = User is authenticated.
- **G** = User is granted access.

Knowledge Base:

1. If the user enters the correct password, then the user is authenticated:

$$P \rightarrow A$$

2. If the user is authenticated, then the user is granted access:

$$A \rightarrow G$$

3. The user entered the correct password:

$$P$$

Goal:

$$G$$

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## 2. Convert to CNF

For the first implication:

$$P \rightarrow A$$

Therefore:

$$\neg P \vee A$$

For the second implication:

$$A \rightarrow G$$

Therefore:

$$\neg A \vee G$$

Fact:

$$P$$

Thus the clauses are:

- **C1:**  $\neg P \vee A$
  - **C2:**  $\neg A \vee G$
  - **C3:**  $P$
- 

## 3. Negate the Goal

Goal:

$$G$$

Negated goal:

$$\neg G$$

Add:

- **C4:**  $\neg G$

Complete clause set:

$$C1 = \neg P \vee A$$

$$C2 = \neg A \vee G$$

$$C3 = P$$

$$C4 = \neg G$$

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## 4. Resolution Steps

### Step 1

Resolve C1 and C3:

$$C1: \neg P \vee A$$

$$C3: P$$

Resolve **P** and  **$\neg P$** .

Result:

$$C5: A$$

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### Step 2

Resolve C2 and C5:

$$C2: \neg A \vee G$$

$$C5: A$$

Resolve **A** and  **$\neg A$** .

Result:

$$C6: G$$

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### Step 3

Resolve C6 and C4:

$$C6: G$$

$$C4: \neg G$$

Resolve **G** and  **$\neg G$** .

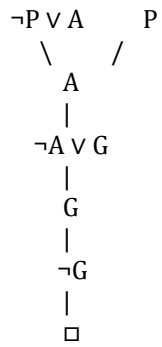
Result:

$$C7: \square$$

The Empty Clause is obtained.

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### Resolution Tree



### 5. Final Conclusion

The resolution process derives:

**$P \rightarrow A \rightarrow G$**

and finally:

**$G + \neg G \rightarrow \square$**

Since the **Empty Clause** ( $\square$ ) is obtained, the goal is proved.

**Therefore, the user is granted access.**

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